

Air Handling Unit Spec Sheet

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Project: Lakefront Office Tower — Phase II

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SECTION 23 73 13 — CENTRAL-STATION AIR HANDLING UNITS

1.0 SCOPE OF WORK

Furnish and install factory-assembled central-station air handling units (AHUs) per drawings M2.01–M2.04. Units serve office floors 3–18, the data center anteroom, and the rooftop mechanical penthouse. N+1 fan redundancy required.

2.0 EQUIPMENT SCHEDULE

Tag	Service	CFM (Design)	ESP (in.w.g.)	Cooling (MBH)	Heating (MBH)	OA %
AHU-1	Low Floors 3–9	28,500	3.50	320	180	25%
AHU-2	High Floors 10–18	26,000	3.75	295	165	25%
AHU-3	Data Anteroom	8,400	2.25	120	60	15%
AHU-4	Penthouse MER	3,200	1.50	48	24	100%

3.0 SYSTEM SCHEMATIC / DRAWING

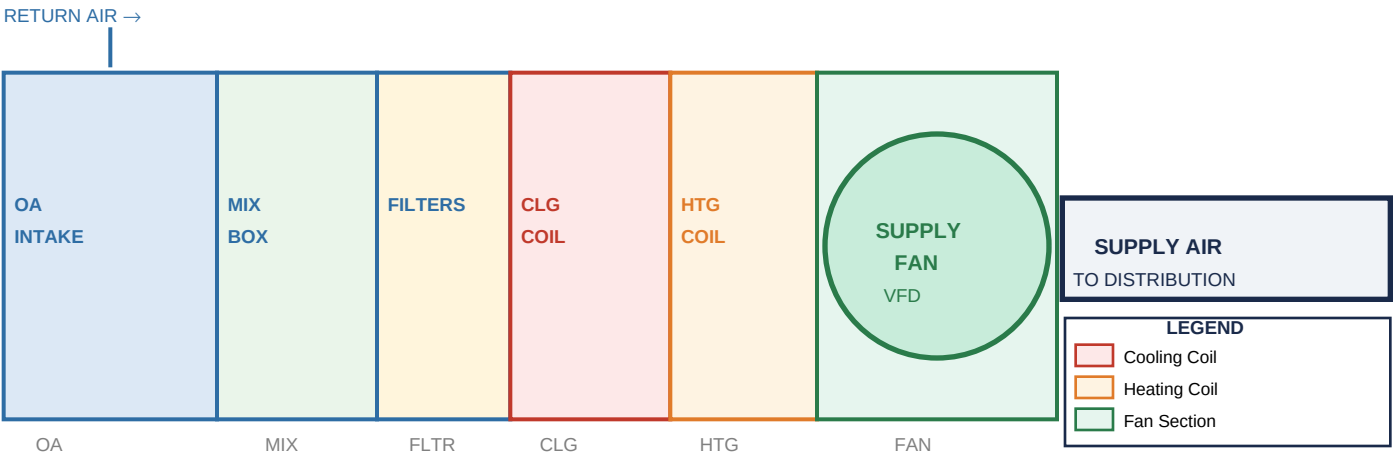


Figure 1 — Typical AHU Section Assembly (Not to Scale)

4.0 TECHNICAL SPECIFICATIONS

4.1 Cabinet & Insulation

Construction	Double-wall, 2-in. insulation (R-13), 20-ga G90 outer / 22-ga inner liner
Thermal Break	Required at all panel joints
Leakage	≤ 1% of design CFM at 1-in. w.g. (ASHRAE 111)
Casing Pressure	–2 to +4 in. w.g. (factory-tested)

4.2 Supply Fan

Fan Type	Airfoil backward-inclined centrifugal
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Drive	VFD — NEMA Premium IE3, 460V/3ph/60Hz
Bearing	L-10 $\geq$ 200,000 hrs (ABMA Std 9)
Vibration	$\leq$ 0.10 in/sec RMS (AMCA 204)
Isolation	2-in. deflection spring + neoprene pad

4.3 Filtration

Pre-Filter	MERV 8, 2-in. pleated — ΔP 0.10/0.50 in. initial/final (ASHRAE 52.2)
Final Filter	MERV 13, 4-in. rigid bag — ΔP 0.25/1.00 in. initial/final (ASHRAE 52.2)

5.0 INSTALLATION NOTES

- AHU installation per SMACNA and manufacturer's written instructions.
- Provide minimum 36-in. service clearance on filter access side.
- Factory run-test each unit at full design conditions; submit certified report.
- AHRI 430 and UL 1995 listings required.
- Coordinate structural curb loads with structural engineer prior to fabrication.

END OF SECTION 23 73 13 — CENTRAL-STATION AIR HANDLING UNITS